Institut für Maschinelle Sprachverarbeitung
Universität Stuttgart
Pfaffenwaldring 5B
D-70569 Stuttgart

Master thesis

Controllable Text Generation Using Langevin Dynamics

Sarthak Singh

Studiengang:	M.Sc. Computational Linguistics
Prüfer*innen:	Prof. Dr. Thang Vu
Betreuer:	Prof. Dr. Thang Vu
Beginn der Arbeit:	05.02.2026
Ende der Arbeit:	31.08.2026

Erklärung (Statement of Authorship)

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbstständig verfasst habe und dabei keine andere als die angegebene Literatur verwendet habe. Alle Zitate und sinngemäßen Entlehnungen sind als solche unter genauer Angabe der Quelle gekennzeichnet. Die eingereichte Arbeit ist weder vollständig noch in wesentlichen Teilen Gegenstand eines anderen Prüfungsverfahrens gewesen. Sie ist weder vollständig noch in Teilen bereits veröffentlicht. Die beigefügte elektronische Version stimmt mit dem Druckexemplar überein.[1]

(Sarthak Singh)

[1] Non-binding translation for convenience: This thesis is the result of my own independent work, and any material from work of others which is used either verbatim or indirectly in the text is credited to the author including details about the exact source in the text. This work has not been part of any other previous examination, neither completely nor in parts. It has neither completely nor partially been published before. The submitted electronic version is identical to this print version.

Inhaltsverzeichnis

1 Introduction

Large Language Models (LLMs) have achieved unprecedented performance in text generation and reasoning. However, despite their fluency, standard autoregressive (AR) decoding provides limited mechanisms for fine-grained, inference-time control. For example, if an autoregressive model generates a fluent but logically inconsistent or toxic phrase, it possesses no mechanism to revise earlier tokens without discarding and regenerating the entire subsequent sequence. Furthermore, enforcing constraints on full sequences is intractable for auto-regressive models, as it requires generating many full sequences in order to pick the one that fullfills the constraints best. While the probabilistic nature of AR decoding is well understood, steering the output to satisfy complex global constraints (such as sentiment, style, or logical consistency) without expensive retraining remains a significant challenge.

Controllable Text Generation (CTG) Hu et al. (2018); Zhang et al. (2023) aims to ensure model outputs adhere to specific constraints while maintaining fluency. Alignment techniques like Reinforcement Learning from Human Feedback (RLHF) can be used to control auto-regressive generation. However, they require a preference dataset and are a ßtatic". More specifically, aligned models cannot easily adapt to new user-defined constraints at runtime without collecting further data that reflects the new preferences and without further fine-tuning. A promising alternative is Energy-Based Model (EBM) guidance, where decoding is treated as an optimization problem. Energy-based decoding methods generate sequences by iteratively refining the whole sequence, often using gradient information as guidance (Langevin Dynamics). Global constraints or guidance signals can be integrated naturally into this framework by adding terms that measure the quality of a sequence to the energy function.

Recently, methods inspired by Langevin Dynamics have been proposed to navigate the text generation landscape. However, many existing approaches adopt Langevin-inspired updates but deviate from classical Stochastic Gradient Langevin Dynamics (SGLD) assumptions Welling and Teh (2011) regarding noise scaling, step-size annealing, or Metropolis-Hastings correction. Consequently, these methods often function more like noisy gradient descentrather than true posterior sampling, potentially compromising the trade-off between control adherence and text diversity.

This thesis proposes to investigate energy-based decoding techniques that allow for the flexible incorporation of sequence-level controls. Specifically, we focus on implementing and evaluating gradient-guided sampling frameworks, such as Langevin Dynamics. By systematically comparing optimization in discrete token spaces versus continuous embedding spaces, we seek to determine if these methods can improve the reliability and semantic coherence of controllable text generation.

2 Literature Review

The field of Controllable Text Generation (CTG) has evolved from simple heuristic-based filtering to complex probabilistic modeling. This section categorizes existing approaches based on the stage of intervention and the nature of the applied constraints, followed by a detailed examination of Langevin Dynamics as the theoretical foundation for this research.

2.1 Taxonomy of Controllable Text Generation

CTG methods can be classified along two primary dimensions: the *stage of intervention* (when the control is applied) and the *granularity of the constraint* (what is being controlled).

2.1.1 Stage of Intervention

Training-based Alignment These methods alter the model parameters θ to inherently bias the model toward desired attributes.

- **Retraining and Supervised Fine-tuning (SFT):** Early approaches involved fine-tuning pre-trained models on attribute-specific datasets Alhafni et al. (2024). However, obtaining high-quality, domain-specific datasets is often difficult and expensive. Furthermore, SFT can lead to mode collapse, where the model generates high-confidence but repetitive text, sacrificing diversity. SFT alone often fails to guarantee strict adherence to complex sequence-level constraints.

- **Reinforcement Learning (RLHF):** Modern alignment techniques use Reinforcement Learning from Human Feedback to optimize a learned reward signal Ouyang et al. (2022); Li et al. (2024). While effective, these methods often suffer from the *alignment tax*, a reduction in output diversity, and are computationally expensive to adapt to new, dynamic constraints at runtime.

Inference-time Guidance (Decoding-time) These methods freeze the model parameters and intervene during the decoding process. This paradigm is the primary focus of this thesis.

- **Prompt Engineering:** Utilizing soft or hard prompts to steer generation Ouyang et al. (2022).
- **Classifier Guidance:** Leveraging external classifiers to adjust token probabilities. Methods such as Plug and Play Language Models (PPLM) Dathathri et al. (2020) and FUDGE Yang and Klein (2021) modify hidden states or predictive distributions based on attribute classifiers.
- **Contrastive Decoding:** Approaches including GeDi Krause et al. (2021) and DExperts Liu et al. (2021) use expert and anti-expert language models to contrastively steer probability mass away from undesirable attributes.

2.1.2 Granularity of Constraints

Constraints in CTG are generally categorized by the scope of their impact on the generated sequence.

Token-Level Constraints These are local constraints applied at specific decoding steps. Examples include hard lexical constraints (forcing inclusion or exclusion of specific words) or syntactic constraints. NeuroLogic Decoding Lu et al. (2021) and K2T Pascual et al. (2021) operate at this level, ensuring the presence of specific keywords in the output.

Global-Level Constraints These constraints apply to holistic properties such as sentiment, topic, or toxicity. Global constraints are difficult to enforce autoregressively because their evaluation often depends on the completed sequence, rendering greedy, next-token prediction insufficient. This thesis focuses primarily on global constraints, where energy-based optimization is most effective.

2.2 Theoretical Foundations: Langevin Dynamics

To address the limitations of autoregressive decoding in satisfying global constraints, recent work has turned to sampling methods rooted in statistical physics and Bayesian inference.

2.2.1 Stochastic Gradient Langevin Dynamics (SGLD)

It is important to note that Langevin Dynamics is inherently defined for continuous distributions. The foundational work of Welling and Teh Welling and Teh (2011) bridges optimization and sampling. In Bayesian learning, one seeks to sample from the posterior distribution

$$p(\theta \mid X) \propto p(X \mid \theta)\, p(\theta). \tag{1}$$

Stochastic Gradient Langevin Dynamics (SGLD) updates parameters according to

$$\Delta\theta_t = \frac{\epsilon_t}{2}\left(\nabla \log p(\theta_t) + \sum_{i=1}^{N} \nabla \log p(x_i \mid \theta_t)\right) + \eta_t, \tag{2}$$

where ϵ_t is the step size and $\eta_t \sim \mathcal{N}(0, \epsilon_t)$ is Gaussian noise.

In the context of controlled generation, the term $\sum_{i=1}^{N} \nabla \log p(x_i \mid \theta_t)$ represents the data likelihood (token probabilities). To adapt this framework for controllable generation, we augment the update rule by adding the gradient of an unnormalized, differentiable constraint function (such as a sentiment classifier), which actively guides the sampling toward the desired attribute.

Welling and Teh showed that by annealing ϵ_t toward zero, the injected noise enables convergence to the true posterior distribution rather than collapsing to a single Maximum A Posteriori (MAP) solution.

However, a major bottleneck in applying this to text is that discrete token spaces do not naturally possess gradients. While Langevin-based solutions work well in continuous spaces, applying them to text requires overcoming this non-differentiable nature, which is a key focus of this research.

2.3 Energy-Based Models for Text

Inspired by SGLD, Energy-Based Models (EBMs) frame text generation as sampling from a distribution defined by an energy function $E(x)$:

$$P(x) \propto \exp(-E(x)), \tag{3}$$

where x denotes a text sequence and lower energy corresponds to better satisfaction of constraints such as fluency or sentiment.

COLD Decoding Qin et al. (2022) and MUCOLA Kumar et al. (2022) apply this framework to large language models by defining the energy as a combination of the model likelihood and differentiable constraint functions. These methods perform iterative updates in the continuous embedding space of the language model.

Research Gap Despite their empirical success, many existing EBM-based methods trade theoretical guarantees for empirical stability. In particular, they often omit the Metropolis–Hastings correction step or fail to properly anneal the noise schedule, resulting in dynamics closer to noisy gradient descent than true sampling. Additionally, the discrete nature of text necessitates ad-hoc de-quantization strategies to reconcile continuous updates with token-level outputs.

This thesis builds upon the Discrete Langevin Sampler proposed by Zhang et al. (2022), which formulates gradient-based proposals directly for discrete distributions. We investigate whether a more mathematically faithful application of Langevin Dynamics can yield improved control and stability in text generation.

2.4 Competitive Landscape: Diffusion Models

Parallel to Langevin-based EBMs, diffusion models have emerged as a non-autoregressive alternative for CTG. DiffuSeq Gong et al. (2023) and EDLM Xu et al. (2025) apply forward and reverse diffusion processes in embedding space. While diffusion and Langevin dynamics are closely related, diffusion models effectively learn the score function used in Langevin sampling; they typically require training separate neural networks to reverse the noise process.

In contrast, the Langevin-based approach explored in this thesis enables zero-shot transfer. It operates directly on the latent space of a frozen, pre-trained language model at inference time, requiring only the definition of an energy function. This offers a more lightweight and flexible mechanism for controllable generation without the need for task-specific parameter updates.

3 Goal and research questions

This thesis investigates contemporary sampling and generation techniques for controlled text generation. The research evaluates how effectively these methods adhere to predefined constraints while maintaining linguistic accuracy. Furthermore, the study explores the internal mechanics of the Llama 3 (8B) model, specifically focusing on its gradient signals and embedding distributions. Through structured experimentation utilizing benchmark datasets and metrics, this work addresses a critical gap identified in the current literature, leading to the following research questions:

RQ1 (Core Methodology): Training-Free Methodological Fidelity and Optimization Space Does enforcing training-free, theoretically correct Langevin noise scaling and annealing improve posterior coverage and stability compared to heuristic approximations?

- **RQ1a** How does the efficiency and text quality of a gradient-based proposal for discrete variables (using embedding gradients) compare to Langevin sampling performed in the continuous embedding space $\mathbf{S} \in R^{L \times D}$?
- **RQ1b** To what extent does the Metropolis–Hastings (MH) correction step mitigate the discretization bias observed in methods such as COLD Qin et al. (2022) or MUCOLA Kumar et al. (2022)?

RQ2 (Application): Robustness and Quality–Control Trade-off Can the proposed rigorous Langevin frameworks satisfy predefined control conditions while maintaining high text quality?

- **RQ2a** How effectively can Langevin-based energy functions satisfy global constraints (e.g., sentiment) while maintaining the standards of fluency and diversity (measured via Perplexity and MAUVE) required for modern LLMs?
- **RQ2b** In the context of gradient-guided editing (text infilling), does energy-based sampling recover semantic coherence more accurately than standard autoregressive decoding?

RQ3: Improving Sampling Efficiency

- **RQ3a** How can we improve upon the efficiency of energy-based methods? Specifically, can we mitigate the computational cost of sampling by incorporating a lightweight training step (e.g., amortized inference or GFlowNet-inspired objectives) to better guide the proposal distribution?

4 Methods

The objective of this research is to move beyond noisy gradient descentänd implement a mathematically sound sampling framework for Controllable Text Generation (CTG). The study will focus on Energy-Based Model (EBM) guidance, where the goal is to sample from a target distribution $p(x) \propto \exp(-E(x))$ defined by an energy function $E(x)$.

4.1 Baseline 1 - Discrete Langevin Sampler (DLS)

To address RQ2, this work will implement the gradient-based proposal for discrete variables proposed by Zhang et al. (2022). Unlike standard Langevin which requires a continuous space, this sampler operates in the discrete token space Θ.

Gradient Approximation Since the token space Θ is discrete, we cannot directly compute the gradient $\nabla U(\theta)$. Instead, we utilize gradients of the energy function with respect to the continuous token embeddings.

We define a local proposal distribution $q(\theta' \mid \theta)$ by scoring candidate tokens based on the inner product between their embeddings and the energy gradient, thereby biasing the selection toward tokens that minimize the energy.

Transition Kernel: The next discrete state θ' is sampled based on the proposal distribution:

$$q(\theta'|\theta) = \frac{\exp(-\frac{1}{2\alpha}||\theta' - \theta - \frac{\alpha}{2}\nabla U(\theta)||_2^2)}{Z_\Theta(\theta)}$$

Normalizing Constant: The term $Z_\Theta(\theta)$ is calculated by summing over the discrete vocabulary space Θ.

Conditioning: To enable controllable generation, the project will incorporate differentiable constraints C (e.g., sentiment scores) into the energy function

$$U(\theta) = -\log p(\theta) + \lambda C(\theta)$$

, where λ is a Lagrange multiplier for the constraint.

Here, Θ can be any space without affecting q being a valid proposal. As a special case, when $\Theta = R^d$, it follows that $Z_{R^d}(\theta) = (2\pi\alpha)^{d/2}$ and recovers the Gaussian proposal in the standard Langevin algorithm. When Θ is a discrete space, we naturally obtain a gradient-based proposal for discrete variables.

Gradient Normalization for Stabilization Early empirical observations revealed that the raw gradients $\nabla U(\theta)$ extracted from the frozen language model exhibit massive variance in magnitude. Because the optimization operates over the log-likelihood of the full sequence, raw gradient steps often act as "rocket thrusters," overshooting the natural language manifold and landing in regions dominated by special or uncalibrated tokens (e.g., <|reserved_special_token|>). To stabilize the proposal distribution, we introduce a gradient normalization step:

$$g_{\text{norm}} = \frac{\nabla U(\theta)}{\nabla U(\theta)_2 + \epsilon}$$

This ensures that the proposal distribution relies purely on the direction of the gradient rather than its magnitude, maintaining the sampler within the safe manifold of natural language embeddings. To maintain a fair comparative baseline, random walk proposals (η) are similarly scaled to unit norm.

4.2 Baseline 2 - Continuous Langevin Sampler (CLS)

This method will address RQ1 and RQ2 by performing optimization in the embedding space of the Large Language Model (LLM). The sequence of embeddings $S \in R^{L \times D}$ will be treated as continuous latent variables.

The methodology involves converting the embedding of the output sequence to a ßoft sequenceänd subsequently optimizing it using Langevin dynamics.

Core Idea. The corrupted segment embedding

$$S \in R^{L \times D}$$

is treated as continuous latent variables, where L is the number of infilling positions and D is the embedding dimension.

Langevin Step (theoretical form).

$$s_{t+1} = s_t + \eta_t \nabla_s \log \pi(s_t) + \sqrt{\eta_t}\, \xi_t, \qquad \xi_t \sim \mathcal{N}(0, I).$$

Equivalent Interpretation. First compute a Gaussian proposal mean

$$m_t = s_t + \eta_t \nabla_s \log \pi(s_t).$$

Then sample

$$s_{t+1} \sim \mathcal{N}(m_t,\, \eta_t I)\,.$$

The Voronoi Cell and "Rubber Band" Effect in CLS While CLS mathematically simulates Langevin dynamics in $R^{L\times D}$, the target density $\pi(s_t)$ fundamentally relies on discrete projections. We define m_s as the interpolation between the continuous step and its discrete projection to prevent infinite drift. However, when gradient normalization is applied to CLS, the continuous step size $\eta_t \nabla_s \log \pi(s_t)$ becomes infinitesimally small relative to the high-dimensional embedding space ($D = 4096$). Consequently, the continuous state fails to escape the Voronoi cell of the current discrete token. The projection step consistently pulls the state back to its origin, resulting in a "rubber band" effect where the sampler flatlines and no token updates occur. Thus, for CLS to function, the raw, unnormalized gradient magnitude is strictly required to forcefully catapult the continuous state across embedding boundaries.

4.3 Metropolis-Hastings (MH) Correction

To address RQ1, we integrate an MH correction step. Standard Langevin-inspired methods (like Qin et al. (2022)) omit this, leading to discretization bias where the sampler may drift away from the true posterior. This involves calculating an acceptance probability:

$$A(\theta'|\theta) = \min\left(1, \frac{p(\theta')q(\theta|\theta')}{p(\theta)q(\theta'|\theta)}\right)$$

This step is intended to restore asymptotic correctness to the target distribution $p(\theta)$, ensuring that the sampler explores the true posterior rather than simply maximizing the objective.

Enforcing Detailed Balance in DLS and CLS To guarantee convergence to the true posterior, the MH acceptance step must strictly enforce detailed balance. For the CLS method, evaluating the target density at the intermediate mean m_s breaks detailed balance. Instead, the forward probability $q(s_{\text{prop}} \mid s_t)$ and the reverse probability $q(s_t \mid s_{\text{prop}})$ must be explicitly calculated using the exact sampled states.

For symmetric random walk baselines, the transition kernels cancel out exactly ($q(s \mid s_{\text{prop}}) = q(s_{\text{prop}} \mid s)$). For the policy-driven methods, the reverse proposal is calculated by evaluating the gradient at the proposed state s_{prop}:

$$m_{\text{prop}} = s_{\text{prop}} + \frac{\epsilon_k}{2}\nabla \log \pi(s_{\text{prop}})$$

The acceptance ratio in log-space is thus defined as:

$$\log \alpha = (\log \pi(s_{\text{prop}}) - \log \pi(s_t)) + (\log \mathcal{N}(s_t \mid m_{\text{prop}}, \epsilon_k) - \log \mathcal{N}(s_{\text{prop}} \mid m_s, \epsilon_k))$$

4.4 Gibbs Sampling for Token Refinement

To provide a comparison for the joint update methods, Gibbs sampling will be implemented. This method updates one token at a time while keeping others fixed:

$$x_i \sim p(x_i|x_{-i}, conditions)$$

The study will evaluate the trade-offs of this approach, specifically comparing its potentially slower convergence against the parallelizable nature of Langevin Dynamics.

4.5 Autoregressive (AR) Decoding Baseline

To answer RQ3, standard Autoregressive (AR) decoding methods, such as Top-P or Beam Search, will be utilized as a performance floor. This baseline represents the model's natural generation without energy-based intervention, allowing for a measurement of the costöf control regarding fluency and diversity.

4.6 The Oracle Step-Size Schedule

To establish an upper bound on the capabilities of Langevin-based decoding and to empirically discover an ideal step-size schedule (α), we introduce an "Oracle" mechanism. The Oracle utilizes the ground-truth (GT) token from the dataset to actively "cheat" during optimization, selecting the step size at each iteration that maximizes the likelihood of transitioning to the GT token. We define a logarithmically spaced grid of candidate step sizes, α_{grid}.

Oracle in the Discrete Langevin Sampler (DLS). In the discrete token space, the Oracle evaluates the unnormalized log proposal probability $q(x_{\text{gt}} \mid \theta_t, \alpha)$ for the ground-truth token x_{gt}. For every optimization step t, we iterate over all $\alpha \in \alpha_{\text{grid}}$ and compute the proposal probability using the gradient-based formulation from Zhang et al. (2022). The log-probabilities are stored, min–max normalized to prevent numerical underflow, and the optimal step size is selected via

$$\alpha^* = \arg\max_{\alpha \in \alpha_{\text{grid}}} q(x_{\text{gt}} \mid \theta_t, \alpha)$$

Once α^* is identified, it is appended to the empirical step-size schedule, and the sampler uses α^* to construct the proposal distribution for sampling the next discrete token.

Oracle in the Continuous Langevin Sampler (CLS). Unlike DLS, the proposal in CLS is not a categorical distribution over the vocabulary, but a multivariate

Gaussian distribution in the continuous embedding space:

$$s_{\text{prop}} \sim \mathcal{N}(m_s(\alpha), \alpha \cdot \text{noise\_scale} \cdot I)$$

where

$$m_s(\alpha) = 0.5 \cdot (\text{interim} + \text{interim\_proj})$$

is our heuristic continuous mean.

To build the Oracle for CLS, we evaluate the Gaussian log-density of the actual ground-truth continuous embedding e_{gt} under the proposal distribution parameterized by each α. The Oracle selects the α^* that maximizes this Gaussian density, effectively choosing the step size that centers the Gaussian proposal as closely over the target GT embedding as possible.

5 Experiment

To systematically evaluate the efficacy of Langevin-based decoding, we conducted an exhaustive ablation study on a flipped token recovery (infilling) task. The experiments were performed using the pre-trained Llama-3 (8B) model on the Wikitext-2 dataset.

5.1 Experimental Setup

We isolate text sequences of 10 to 60 words to represent standard sentence lengths. For each sequence, M tokens, where $M \in \{1, 2, 3\}$, are randomly selected (excluding BOS/EOS tokens) and replaced with uniformly sampled random tokens from the vocabulary, yielding a corrupted sequence.

The sampler is then tasked with recovering the semantic coherence of the sequence over $K = 50$ Langevin steps. The continuous step-size ϵ_k is annealed linearly.

We evaluate three distinct directional policies:

- **DLP Policy:** Following the exact gradient of the energy function.

- **Grad Norm Preserved Random:** A stochastic baseline where the direction is random, but the step size (norm) exactly matches the policy gradient.
- **Pure Random:** A standard Gaussian random walk $\mathcal{N}(0, I)$.

5.2 The Metric Conflict: Anisotropy in LLM Embeddings

Initially, performance was measured simultaneously via L2 Euclidean Distance, comparing the proposed token embedding to the ground-truth token embedding, and KL Divergence, measuring the difference in the autoregressive predictive distribution $P(x_{t+1} \mid x_{\leq t})$.

Findings. We observed a stark contradiction between these metrics. Methods often achieved excellent L2 distances while producing catastrophic KL divergence scores, and vice versa.

Analysis. This divergence is symptomatic of the well-documented anisotropy in LLM embedding spaces. High-dimensional token embeddings are typically clustered in narrow cones. Consequently, a token that is mathematically "close" in L2 space might represent a severe grammatical violation (high KL). Conversely, the sampler may find a grammatically perfect synonym that lies far away in L2 space (low KL, high L2).

Conclusion. Moving forward, L2 distance was deemed an unreliable proxy for semantic quality. KL Divergence (contextual fit) and Entropy (prediction uncertainty) were adopted as the primary arbiters of generation quality.

> [**Insert Plot Here**] Add the bar charts or scatter plot showing the trade-off between L2 Distance and KL Divergence, where L2 is low but KL is high for certain methods.

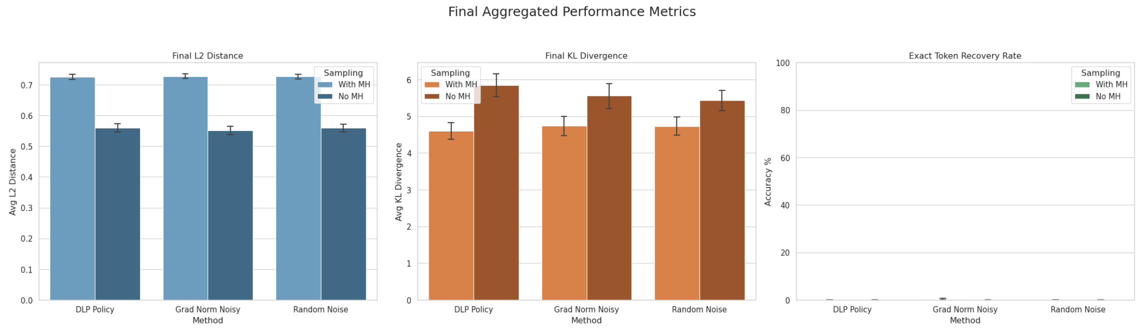

Abbildung 1: Final aggregated metrics for Discrete Langevin Sampling. Note the contradiction: the No MHconfigurations achieve lower L2 Euclidean distance but suffer from significantly higher KL Divergence, highlighting the anisotropy of the embedding space.

Limitations of Step-wise KL Divergence. While KL Divergence provided a better proxy for semantic coherence than Euclidean distance, theoretical discussions during this research revealed a fundamental limitation in its step-wise application $P(x_{t+1} \mid x_{\leq t})$. The magnitude of the divergence is heavily dependent on the absolute position of the corrupted token. Masking an early token alters the conditional probability of the entire subsequent sequence, yielding a massive divergence score, whereas masking the final token yields a near-zero divergence regardless of the token's quality.

A theoretically pure evaluation would require running the sampler thousands of times to form an empirical distribution for the masked token, to compare directly against the model's logits. This computational intractability further motivates the need for amortized inference models that can evaluate the joint probability of the entire generated sequence at once.

5.3 The Oracle Step-Size Schedule vs. Free-Running Dynamics

The primary goal of the Oracle experiment was twofold: (1) to discover the ideal empirical step-size schedule for Langevin-based token recovery, and (2) to establish a performance ceiling for gradient-guided text generation.

Building upon this Oracle framework, we systematically introduced several modifications to analyze the sampler's behavior:

- **Gradient Information Experiment:** Replacing the true gradient with a norm-preserved random vector to ablate the directional value of the LLM's gradient.
- **Temperature Sampling Modification:** Applying a temperature scaling factor to the logits prior to softmax to control the entropy of the proposal distribution.
- **Entropy Plotting:** Tracking the Shannon entropy of the proposal distribution at each step to visualize the sampler's predictive uncertainty over time.

5.3.1 Expected vs. Actual Oracle Behavior

Theoretically, if the gradient vector perfectly encoded the direction to the optimal token, one would expect the Oracle to exhibit a "one-and-done" schedule: selecting a massive α at step 1 to jump directly to the ground-truth token, followed by $\alpha = 0$ for all remaining steps.

However, our empirical analysis of the generated α-schedule revealed a starkly different reality. The Oracle did not take one massive step; instead, it selected a sequence of smaller, sustained step sizes over multiple iterations.

Analysis of the Schedule. This phenomenon occurs because the gradient $\nabla U(\theta)$ provided by the language model is merely a local, linear approximation of a highly non-linear, non-smooth energy landscape. A massive step ($\alpha \to \infty$) in the direction of the local gradient does not point directly to the ground-truth token; instead, due to the severe anisotropy and curvature of the LLM embedding space, a massive step overshoots the natural language manifold entirely. Consequently, the Oracle is forced to select smaller α values, iteratively updating the local gradient to navigate the curved manifold toward the ground-truth embedding.

An initial data leakage issue was discovered in early random-gradient experiments, where the initialization inadvertently included the ground-truth embedding plus noise. Because the proposal equations balance direction against squared distance, the Oracle artificially selected small α values to remain near the leaked ground-truth embedding. Once this issue was resolved, the necessity of iterative, guided stepping was confirmed.

5.3.2 Free-Running Dynamics (Oracle OFF)

To simulate real-world inference, where the ground truth is unknown, we deactivated the Oracle, forcing the models to rely on a fixed, linearly decaying step-size schedule.

Findings. When the Oracle was deactivated, exact token recovery accuracy naturally decreased. However, the KL divergence improved significantly.

Analysis. By forcing the model to step toward the dataset's specific ground-truth token, the Oracle was actively fighting the model's internal autoregressive probability distribution. In natural language, there are many valid ways to complete a sentence. When allowed to run freely (Oracle OFF), the Langevin gradients successfully healed the corrupted text by drifting toward the most mathematically likely contextual fit, often a valid synonym or grammatically correct alternative, proving that the core mechanics of the gradient signals were functioning.

> [**Insert Plot Here**] Add the line plot showing the empirically discovered α-schedule over time, to visualize how it did not simply drop to zero after step 1.

> [**Insert Plot Here**] Add the trajectory line plot showing KL divergence dropping over time for the "Oracle OFF" experiments.

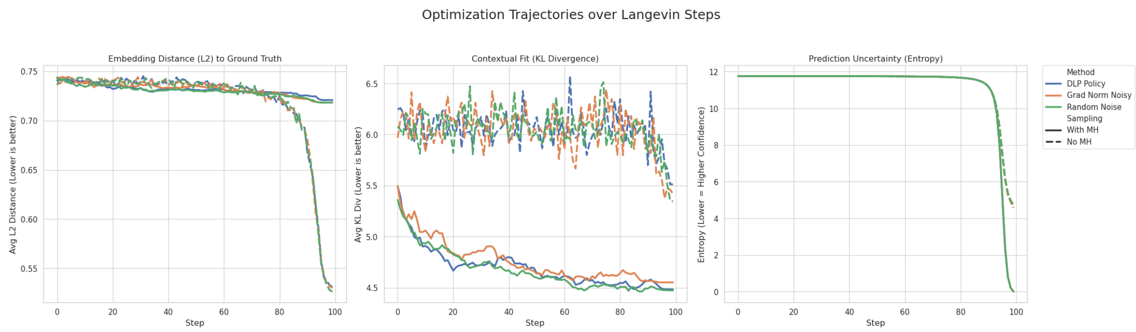

Abbildung 2: When the Oracle is deactivated, the algorithm relies entirely on the autoregressive distribution. Strikingly, the Random Noise"baseline matches the performance of the exact "DLP Policy", demonstrating that the gradient's direction is largely uninformative in discrete text generation.

5.4 Annealing Dynamics and the Quenching Effect in DLS

Note. The phenomenological findings detailed in the subsequent sections, specifically regarding annealing dynamics, Metropolis-Hastings breakdowns, and gradient fallacy, were observed to be consistent across both single-token ($M = 1$) and multi-token ($M > 1$) recovery tasks. As the number of coupled masked tokens increased, the absolute recovery accuracy naturally degraded, but the relative behavioral differences between the sampling algorithms remained strictly preserved.

During the evaluation of the Discrete Langevin Sampler (DLS) under free-running dynamics (Oracle OFF), a peculiar artifact was observed: in a standard 50-step optimization schedule, configurations lacking Metropolis–Hastings (MH) sampling exhibited highly chaotic metrics for the majority of the run, followed by a sudden, precipitous "crash" in L2 distance, KL divergence, and entropy around step 40.

To determine whether this crash was a result of the model requiring more steps to reach convergence, or an artifact of the step limit itself, we designed a follow-up experiment extending the DLS optimization to 100 steps.

Findings. The 100-step experiment revealed that the sudden drop in metrics did not remain at step 40; rather, it shifted entirely to the final 15 steps of the prolonged schedule (steps 85–100). Conversely, the configurations utilizing the MH correction step (solid lines) ignored this phenomenon entirely, converging smoothly and rapidly within the first 20 steps and maintaining a stable, low KL divergence for the remainder of the schedule.

Analysis of the Quenching Effect. The shifting of the metric crash proves that this behavior is entirely governed by the linear annealing schedule of the step-size ϵ_k. The configurations without MH lack a mathematical rejection mechanism; consequently, they accept all proposals regardless of their impact on the sequence's joint probability.

The "Boiling" Phase. During the first 80% of the schedule, ϵ_k is sufficiently large that the injected Gaussian noise dominates the proposal. The "No MH" sampler bounces stochastically across the discrete vocabulary, exhibiting maximum predictive uncertainty (high entropy) and poor contextual fit (high KL divergence).

The "Quenching" Phase. As the optimization reaches the end of its schedule ($k \to K$), ϵ_k decays to its minimum bound ($\epsilon_{\min}$). The stochastic noise effectively vanishes, transitioning the algorithm into deterministic gradient descent. Stripped of noise, the sampler greedily collapses into the nearest local minimum, causing the sudden plummet in metrics.

This dynamic sharply contrasts with the "With MH" configurations. The MH acceptance ratio acts as a strict probabilistic gatekeeper. Rather than blindly accepting the high-temperature noise, MH rejects proposals that egregiously harm the sequence likelihood. This allows the sampler to safely navigate down the energy landscape early in the schedule.

Crucially, even after the "No MH" methods experience their late-stage quenching phase, their final KL divergence remains strictly worse than the "With MH" counterparts. This confirms that relying on late-stage cooling without detailed balance traps the sampler in suboptimal local minima. We conclude that extending the optimization schedule beyond 50 steps is computationally wasteful; 30 to 50 steps, equipped with MH sampling, provides the optimal balance of rapid convergence and high-fidelity token recovery.

> [**Insert Plots Here**] Insert both the 50-step DLS plot and the 100-step DLS plot side-by-side or stacked. This provides direct visual evidence of the crash shifting from step 40 to step 85.

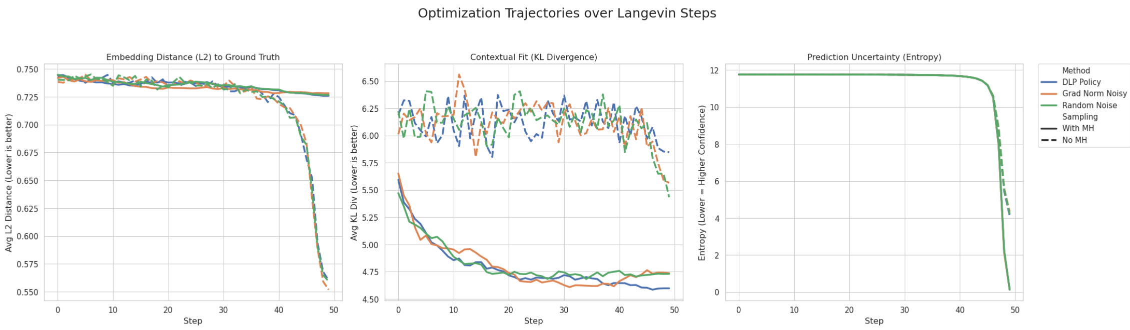

Abbildung 3: Figure Z(a): DLS trajectory over 50 steps. The No MHconfigurations (dashed lines) experience a sudden crash in metrics near step 40 as the noise schedule approaches zero

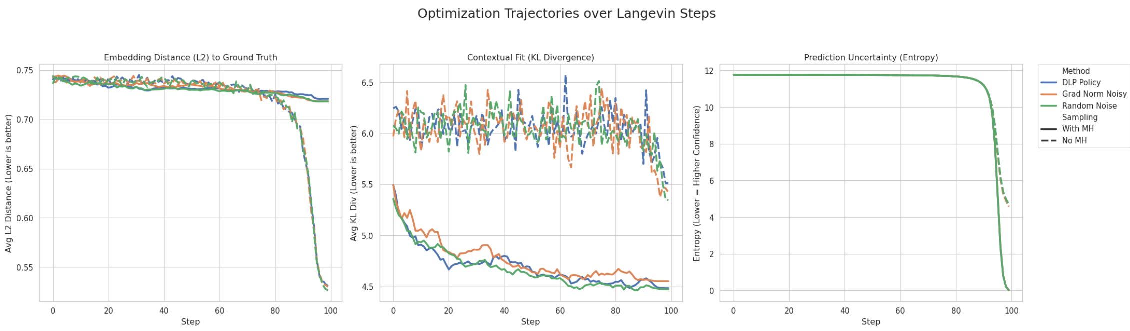

Abbildung 4: Figure Z(b): DLS trajectory over 100 steps. The metric crash shifts entirely to step 85, confirming that this phenomenon is driven by the quenching of the linear noise schedule ($\epsilon_k \to 0$) rather than actual model convergence.

5.5 The Metropolis–Hastings Breakdown in Continuous Space (CLS)

The most profound discovery of this thesis occurred when comparing the Metropolis–Hastings (MH) acceptance dynamics between the Discrete Langevin Sampler (DLS) and the Continuous Langevin Sampler (CLS).

Mathematically, MH is required to correct discretization bias and ensure the sampler adheres to the target distribution. In DLS, enabling MH functioned perfectly: it acted as a strict grammar gatekeeper, rejecting stochastically poor proposals and strictly lowering the final KL divergence compared to DLS without MH.

However, in CLS, enabling MH resulted in catastrophic failure. The sampler flatlined, and the MH step rejected almost 100% of the proposals.

Theoretical Analysis. This failure highlights a fundamental incompatibility between continuous MCMC methods and discrete text representations.

In CLS, the continuous state s operates in R^D. However, the energy function $U(s) = -\log \pi(s)$ relies on a nearest-neighbor projection into the discrete vocabulary. This creates a landscape of Voronoi cells.

The target density surface is not a smooth, differentiable hill, but rather a series of perfectly flat plateaus separated by massive, discontinuous cliffs.

When CLS proposes a step s_{prop} that crosses a cell boundary into a new token's territory, the sequence probability $\pi(s)$ instantly experiences a massive, discontinuous jump. The MH algorithm evaluates this jagged jump, assumes it is a mathematically invalid leap in the energy landscape, and rejects it.

Conclusion. While CLS without MH can blindly force its way across these boundaries to find better tokens, adding theoretically rigorous MH corrections paralyzes the sampler. Continuous Langevin dynamics fundamentally break down on step-function energy landscapes.

[Insert Plot Here] Add the trajectory line plots comparing CLS, where metrics are flat or stuck, against DLS, where metrics smoothly converge. Also add the bar chart showing `mh_enabled=True` hurting CLS but helping DLS.

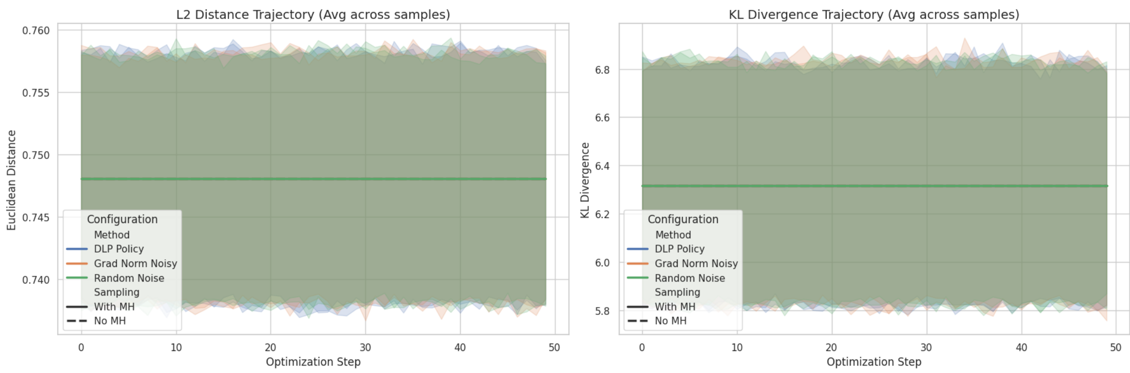

Abbildung 5: Figure W(a): Continuous Langevin Sampler with Gradient Normalization enabled. The infinitesimally small step sizes fail to escape the token's Voronoi cell, resulting in a rubber bandëffect where optimization completely flatlines.

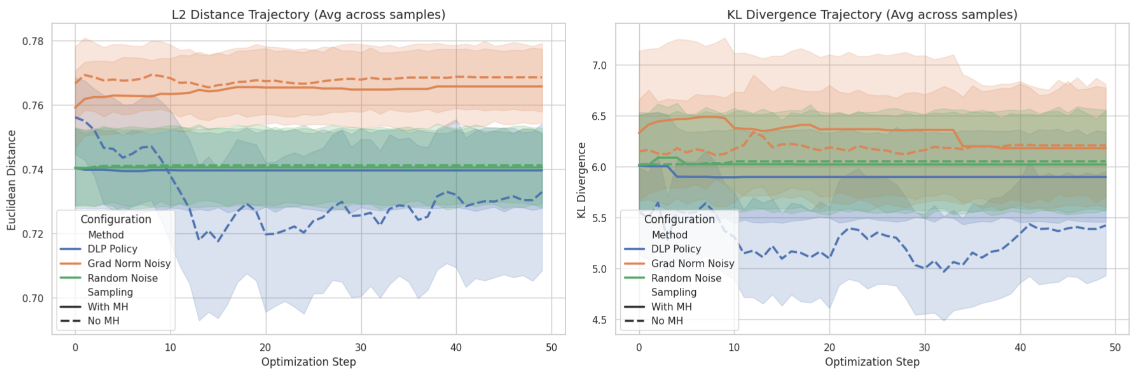

Abbildung 6: Figure W(b): CLS with Gradient Normalization disabled. Unrestricted step sizes allow the No MHconfigurations (dashed lines) to escape local boundaries and reduce KL divergence. However, the Metropolis-Hastings correction (solid lines) consistently rejects these discontinuous jumps across the energy landscape, paralyzing the sampler.

6 Discussion and Future Work

6.1 Synthesizing the Limits of Gradient-Guided Decoding

The primary objective of this thesis (RQ1 and RQ2) was to determine whether rigorous, training-free Langevin dynamics could effectively control large language models at inference time.

Through extensive experimentation, we conclude that gradient-based sampling on pre-trained LLMs is fundamentally limited due to the nature of their training objective.

Standard autoregressive LLMs (such as Llama-3) are trained via teacher forcing to minimize local, token-level conditional cross-entropy. They are not trained as global energy-based models (EBMs). Consequently, they do not possess a smooth, well-calibrated global energy landscape over full sequences.

When we extract the gradient $\nabla \log p(x)$ to guide our sampler, we are relying on an uninformative, highly jagged signal. The gradient points blindly, often shooting the continuous state out of the natural language distribution entirely. While gradient normalization in DLS mitigates catastrophic divergence, the underlying issue remains: the energy function of a standard LLM is too poorly behaved to support reliable, high-fidelity MCMC sampling without extensive heuristic adjustment.

The Fallacy of the Gradient Direction A cornerstone of our investigation was the "Gradient Information Experiment," designed to answer RQ1: how much information is contained in the gradient-following policy? We hypothesized that substituting the LLM's exact gradient with a random Gaussian vector would drastically degrade token recovery. Crucially, to ensure a mathematically fair comparison, the random vector had to be strictly normalized. In a high-dimensional embedding space $D = 4096$, a standard normal Gaussian vector $\mathcal{N}(0, I)$ possesses an expected norm of approximately $\sqrt{D} \approx 64$.

Without strict ℓ_2-normalization, a random baseline would take optimization steps orders of magnitude larger than the normalized policy gradient, breaking detailed balance. By forcing both the policy gradient and the random direction to an exact unit norm, we isolated the effect of the gradient's direction.

Counterintuitively, the results demonstrated that the random-direction baseline performed remarkably well, often recovering the correct token or achieving comparable KL divergence to the true policy. This finding implies that the direction of the autoregressive gradient in discrete text generation is highly uninformative. Because the energy landscape is composed of jagged, discontinuous Voronoi cells rather than a smooth continuous surface, the exact gradient vector offers little advantage over a scaled random walk. The primary driver of optimization is not the gradient direction itself, but rather the magnitude of the step and the subsequent accept–reject evaluation. This realization directly motivates our proposed future work: abandoning gradient-based proposals in favor of trained, amortized search policies such as GFlowNets.

6.2 Future Work: Amortized Inference via GFlowNets

Given the theoretical limits of gradient-based proposals discovered in this work, future efforts to answer RQ3, improving sampling efficiency and control, must pivot away from gradient-reliant methods.

If the LLM's gradient is uninformative, we must instead rely purely on the model's ability to rank the quality of sequences through its absolute probability output. In the GFlowNet framework, the frozen autoregressive LLM is repurposed strictly as an unnormalized reward function, $R(x) = \exp(-E(x)) = p(x)$. The auxiliary GFlowNet policy is trained via Trajectory Balance (TB) or Detailed Balance (DB) objectives to explore the discrete sequence space. Rather than relying on jagged local derivatives to navigate token boundaries, the GFlowNet amortizes the search process, learning to sample trajectories such that the marginal probability of generating a sequence is directly proportional to the global reward assigned by the LLM.

A highly promising direction is provided by Generative Flow Networks (GFlowNets). Unlike Langevin dynamics, which attempt to navigate the jagged energy landscape using local derivatives, GFlowNets reframe generation as a reinforcement learning problem. An auxiliary, lightweight policy network is trained to construct sequences sequentially such that the marginal probability of sampling a sequence is exactly proportional to its reward, as evaluated by the frozen LLM.

Recent work on intractable inference suggests that GFlowNets can successfully explore discrete, discontinuous spaces where MCMC methods suffer from mixing issues or become trapped by boundary cliffs, which is precisely the failure mode observed in our CLS experiments. By training a GFlowNet to amortize the search process, we can bypass the broken gradient signals of the LLM while still leveraging its world knowledge to achieve highly fluent, controllable text generation.

Literatur

Bashar Alhafni, Vivek Kulkarni, Dhruv Kumar, and Vipul Raheja. Personalized text generation with fine-grained linguistic control, 2024. URL https://arxiv.org/abs/2402.04914.

Sumanth Dathathri, Andrea Madotto, Janice Lan, Jane Hung, Eric Frank, Piero Molino, Jason Yosinski, and Rosanne Liu. Plug and play language models: A simple approach to controlled text generation, 2020. URL https://arxiv.org/abs/1912.02164.

Shansan Gong, Mukai Li, Jiangtao Feng, Zhiyong Wu, and Lingpeng Kong. Diffuseq: Sequence to sequence text generation with diffusion models, 2023. URL https://arxiv.org/abs/2210.08933.

Zhiting Hu, Zichao Yang, Xiaodan Liang, Ruslan Salakhutdinov, and Eric P. Xing. Toward controlled generation of text, 2018. URL https://arxiv.org/abs/1703.00955.

Ben Krause, Akhilesh Deepak Gotmare, Bryan McCann, Nitish Shirish Keskar, Shafiq Joty, Richard Socher, and Nazneen Fatema Rajani. GeDi: Generative discriminator guided sequence generation. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 4929–4952, Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-emnlp.424. URL `https://aclanthology.org/2021.findings-emnlp.424/`.

Sachin Kumar, Biswajit Paria, and Yulia Tsvetkov. Gradient-based constrained sampling from language models, 2022. URL `https://arxiv.org/abs/2205.12558`.

Wendi Li, Wei Wei, Kaihe Xu, Wenfeng Xie, Dangyang Chen, and Yu Cheng. Reinforcement learning with token-level feedback for controllable text generation, 2024. URL `https://arxiv.org/abs/2403.11558`.

Alisa Liu, Maarten Sap, Ximing Lu, Swabha Swayamdipta, Chandra Bhagavatula, Noah A. Smith, and Yejin Choi. DExperts: Decoding-time controlled text generation with experts and anti-experts. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 6691–6706, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.522. URL `https://aclanthology.org/2021.acl-long.522/`.

Ximing Lu, Peter West, Rowan Zellers, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. NeuroLogic decoding: (un)supervised neural text generation with predicate logic constraints. In Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou, editors, *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4288–4299, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.339. URL `https://aclanthology.org/2021.naacl-main.339/`.

Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback, 2022. URL https://arxiv.org/abs/2203.02155.

Damian Pascual, Beni Egressy, Clara Meister, Ryan Cotterell, and Roger Wattenhofer. A plug-and-play method for controlled text generation. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3973–3997, Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-emnlp.334. URL https://aclanthology.org/2021.findings-emnlp.334/.

Lianhui Qin, Sean Welleck, Daniel Khashabi, and Yejin Choi. Cold decoding: energy-based constrained text generation with langevin dynamics. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088.

Max Welling and Yee Whye Teh. Bayesian learning via stochastic gradient langevin dynamics. In *Proceedings of the 28th International Conference on International Conference on Machine Learning*, ICML'11, page 681–688, Madison, WI, USA, 2011. Omnipress. ISBN 9781450306195.

Minkai Xu, Tomas Geffner, Karsten Kreis, Weili Nie, Yilun Xu, Jure Leskovec, Stefano Ermon, and Arash Vahdat. Energy-based diffusion language models for text generation. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 33769–33789, 2025. URL https://proceedings.iclr.cc/paper$_{f}iles/paper/2025/file/535baca36883a4e0450423b26b150a4$
$Paper - Conference.pdf$.

Kevin Yang and Dan Klein. Fudge: Controlled text generation with future discriminators. In *Proceedings of the 2021 Conference of the North*

American Chapter of the Association for Computational Linguistics: Human Language Technologies, page 3511–3535. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.naacl-main.276. URL `http://dx.doi.org/10.18653/v1/2021.naacl-main.276`.

Hanqing Zhang, Haolin Song, Shaoyu Li, Ming Zhou, and Dawei Song. A survey of controllable text generation using transformer-based pre-trained language models, 2023. URL `https://arxiv.org/abs/2201.05337`.

Ruqi Zhang, Xingchao Liu, and Qiang Liu. A langevin-like sampler for discrete distributions, 2022. URL `https://arxiv.org/abs/2206.09914`.